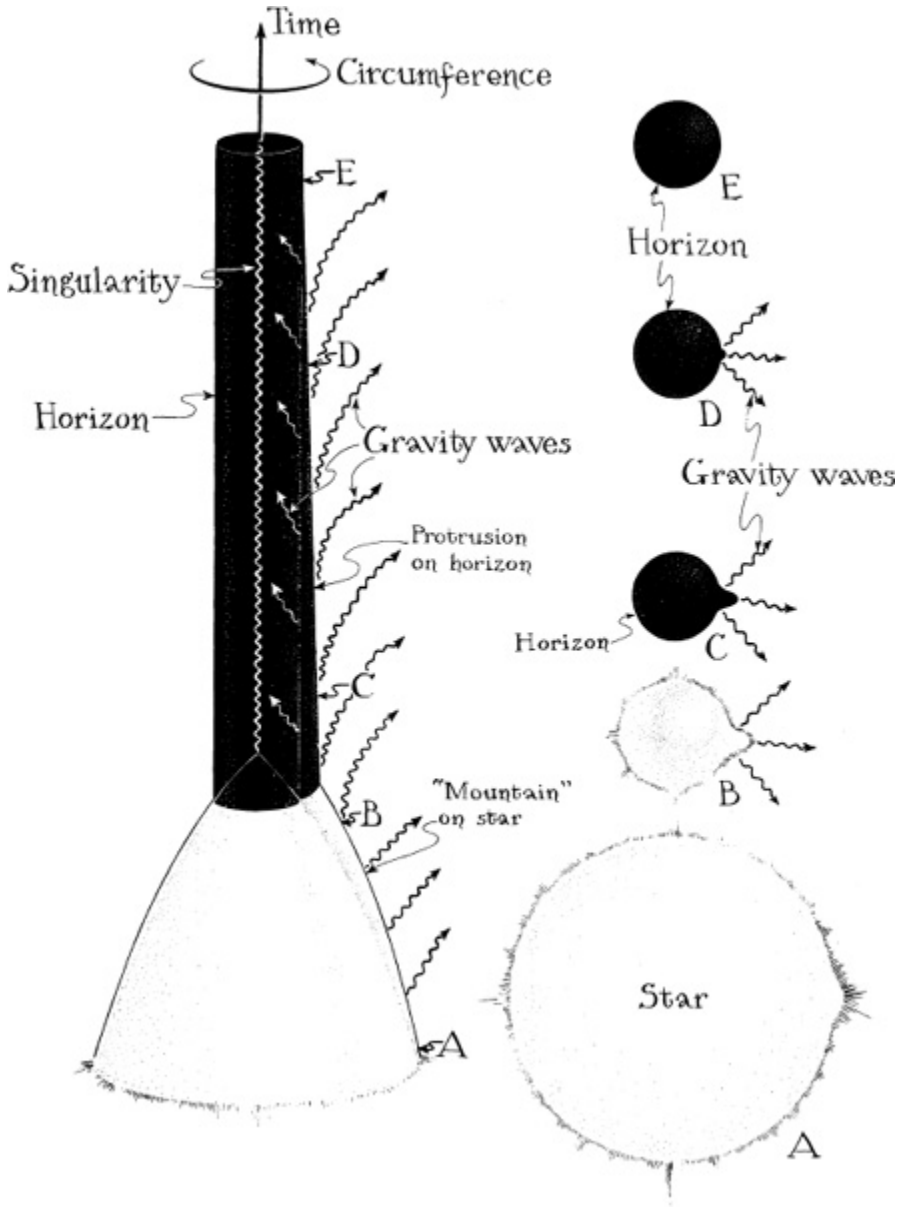
fly out into the surrounding Universe, and as they fly away, the ripples leave the hole with a perfectly spherical shape



7.4 Spacetime diagram (*left*) and a sequence of snapshots *(right)* showing the implosion of a mountain-endowed star to form a black hole.

A familiar analogue is the plucking of a violin string. So long as one's finger holds the string in a deformed shape, it remains deformed; so long as the mountain is protruding out of the hole, it keeps the newborn horizon deformed. When one removes one's finger from the string, the string vibrates, sending sound waves out into the room; the sound waves carry away the energy of the string's deformation, and the string settles down into